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FACULTEIT INDUSTRIËLE INGENIEURSWETENSCHAPPEN

CAMPUS GROEP T LEUVEN

Master's thesis title

Subtitle (optional)

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Write a short summary in Dutch that is accessible to a wide audience and contains a **maximum of 3500 characters**. This summary must also be uploaded to KU Loket.

Summary

This (short) summary, accessible to a wide audience, is written in English and contains a **maximum of 3500 characters**. This summary must also be uploaded to KU Loket.

Keywords: Enter (more or less) five keywords.

Inhoudsopgave

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1.1 Example text 1

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Acroniemen

IoT Internet of Things. 2

LPWAN low-power wide-area network. 2

Hoofdstuk 1

Changes w.r.t. the default template

1.1 Example text

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1.2 Template features

- Changed font from default computer modern to CMBright.
- Increased line spacing in the title to prevent letters from overlapping.
- The siunitx package is included; see examples at: `https://github.com/dramco-edu/LaTex/wiki/LaTex-Tips-and-Tricks#using-si-units-numbers-angles-etc`
- Include units alongside equations using the following command:

```
\begin{equation}
 E = m c^2
 \tagaddtext{[\si{\joule}]}
\end{equation}
```

  , which outputs:

$$E = mc^2 \qquad \text{[J]} \quad (1.1)$$

- Abbreviations:
  - Internet of Things (IoT)
  - low-power wide-area networks (LPWANs)
  - Internet of Things
  - IoT
  - Internet of Things (IoT)
- `\ce{CO2}` CO_2
- Refs:
  - Figuur 1.1
  - Tabel 1.1
  - Hoofdstuk 1
  - Vergelijking (1.1)
  - `\cref{fig:example}` becomes Figuur 1.1

  - `\cref{tab:example}` becomes Tabel 1.1
  - `\cref{ch:introduction}` becomes Hoofdstuk 1
  - Vergelijking (1.1)

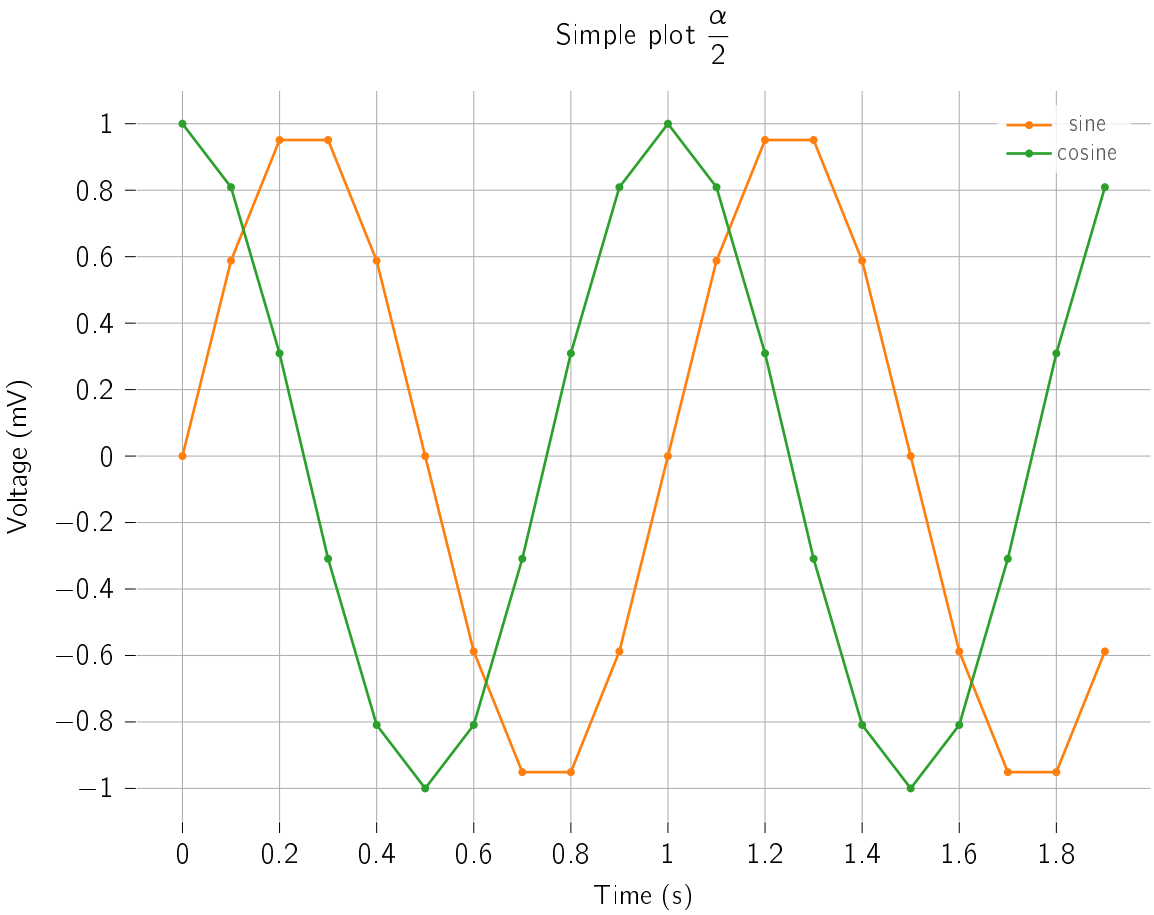
- numerical citation [1]
  - `\cite{callebaut10lorawan}` becomes [2]
  - `\citeauthor{callebaut10lorawan}` becomes Callebaut
  - `\citeauthorref{callebaut10lorawan}` becomes Callebaut [2]
  - `\autocite[chap.~2]{callebaut10lorawan}` becomes [2, chap. 2]
  - `\footcite{callebaut10lorawan}` becomes[1]
  - `\fullcite{callebaut10lorawan}` becomes G. Callebaut, "LoRaWAN Network Simulator", *doi*, jrg. 10, p. 5281,

- A 45° angle or a 45°.
- It is 17 °C outside.
- 10 000
- 3.45×10^{-4}
- $\mathrm{kg\,m\,s^{-2}}$
- $\mathrm{kg\,m/s^2}$
- 10 %
- 68 kg

[1] 2.



Figuur 1.1: Example JPG



Figuur 1.2: Caption

Day	Min Temp	Max Temp	Summary
Monday	11C	22C	A clear day with lots of sunshine. However, the strong breeze will bring down the temperatures.
Tuesday	9C	19C	Cloudy with rain, across many northern regions. Clear spells across most of Scotland and Northern Ireland, but rain reaching the far northwest.
Wednesday	10C	21C	Rain will still linger for the morning. Conditions will improve by early afternoon and continue throughout the evening.

Day	Min Temp	Max Temp	Summary
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Wednesday	10C	21C	Rain will still linger for the morning. Conditions will improve by early afternoon and continue throughout the evening.

Tabel 1.1 Fixed-width columns.

	Treatment A	Treatment B
John Smith	Good response, no side-effects	No response
Jane Doe	–	Good response, no side-effects
Mary Johnson	No response	Good response with side-effects

```
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M, VT), 1)

                VT = np.zeros((n*m,1), int)

    return M
```

Listing 1: Minimal working example

Bibliografie

[1] G. Ottoy, G. Leenders en G. Callebaut, “LoRaWAN EFM32”,

[2] G. Callebaut, “LoRaWAN Network Simulator”, *doi*, jrg. 10, p. 5281,

Bijlage A

An appendix

Add supporting material here.

Bijlage B

Scientific article describing this thesis

The thesis should also contain a short scientific article. If you write your thesis in Dutch, you must write the article in English, and vice versa. We advise you to employ the IEEE Manuscript Templates for Conference Proceedings (`https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn`). Compile the article in another project and include the generated pdf file as shown below:

Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

5th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

6th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

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This document is a model and instructions for LaTeX. Please observe the conference page limits.

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III. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

Identify applicable funding agency here. If none, delete this.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m^2” or “webers per square meter”, not “webers/m^2”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm^3”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not

Bijlage C

Poster

Bijlage D

GenAI code of conduct

Placeholder

GenAI code of conduct

Download the most recent KU Leuven GenAI code of conduct, fill it in, export the completed document to PDF, and manually include that PDF here.

The files `chapters/GenAI_Code of conduct_AJ 2526_*` are provided as a starting point only. Always check the KU Leuven website for the most recent version before submitting your thesis.

Replace this placeholder with, for example:

```
\includepdf [pages=-]{chapters/code-of-conduct.pdf}
```